WEAKLY ASSOCIATED POINTS OF A PUSHFORWARD ALONG AN AFFINE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The proof is a reduction to algebra: the affine translation is done in place and the algebra content is delegated to the cited Algebra lemma; that delegation is recorded here for the reader.

Editorial note. This card is used as an input by the card for lemma-ass-functorial-equal (L44211), where it enters as a navigation pointer.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Weakly associated points of a sheaf on an affine scheme are the weakly associated primes. Lemma lemma-weakly-associated-affine-open, used at proof lines 823-829. Translates both sides into algebra.
- The reverse functoriality of weakly associated primes under a ring map. Algebra, Lemma algebra-lemma-weakly-ass-reverse-functorial, used at proof lines 829-831. Is the algebra statement that finishes the proof.

Source reference. Original lemma 05C6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6909–6909. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 056M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6715–6715. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to S$ be an affine morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then we have*

$$\mathit{WeakAss}_S(f_*\mathcal{F}) \subset f(\mathit{WeakAss}_X(\mathcal{F}))$$

Proof. We may assume X and S affine, so $X \to S$ comes from a ring map $A \to B$. Then $\mathcal{F} = \widetilde{M}$ for some B-module M. By Lemma 056M the weakly associated points of $\mathcal{F}$ correspond exactly to the weakly associated primes of M. Similarly, the weakly associated points of $f_*\mathcal{F}$ correspond exactly to the weakly associated primes of M as an A-module. Hence the lemma follows from Algebra, Lemma 05C6. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.